



Mark Scheme (Results)

November 2025

Pearson Edexcel International GCSE in Chemistry (Modular)
4WCH2/1C

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November 2025

Question Paper Log Number P81560RA

Publication Code 4WCH2_1C_2511_MS

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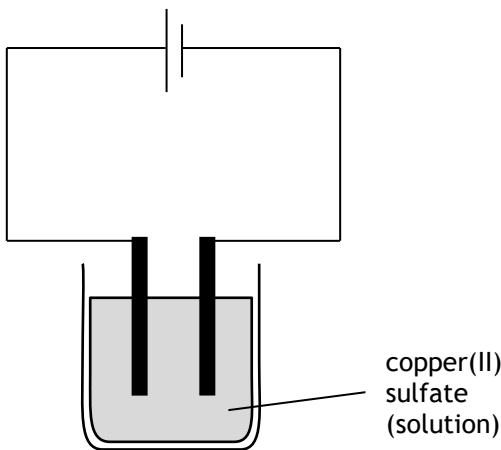
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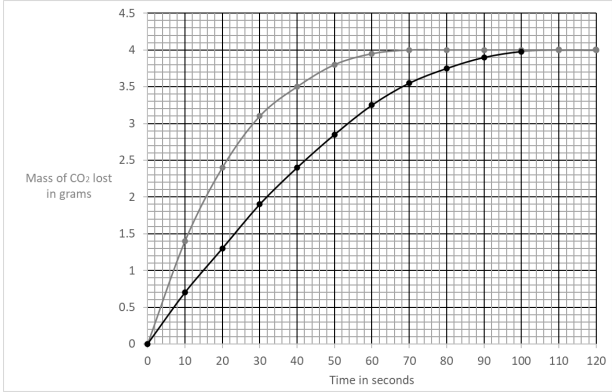
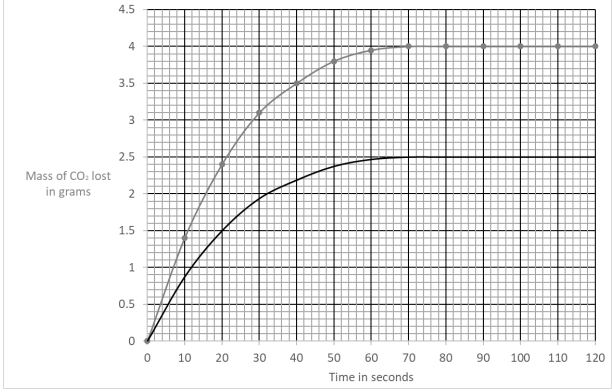
General Marking Guidance

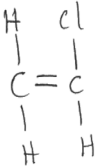
- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

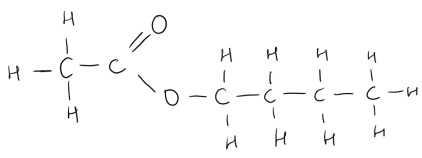
Question number	Answer	Notes	Marks
1 (a)	Iron ion Colour of the precipitate	REJECT multiple lines from left-hand boxes	2
(b)	M1 (litmus paper) turns blue (in presence of the gas) M2 (because gas is) alkaline / basic M3 ammonia/NH₃	ACCEPT idea that gas produces hydroxide (ions)	3
			Total 5

Question number	Answer	Notes	Marks
3 (a)	B (chlorine) <i>A bromine is incorrect as it is not pale green</i> <i>C fluorine is incorrect as it is not pale green</i> <i>D iodine is incorrect as it is not pale green</i>		1
(b)	C (fluorine) <i>A astatine is incorrect as it is less reactive than chlorine</i> <i>B bromine is incorrect as it is less reactive than chlorine</i> <i>D iodine is incorrect as it is less reactive than chlorine</i>		1
(c) (i)	simple molecular		1
(ii)	M1 bromine has a smaller relative molecular mass M2 (so) bromine has weaker intermolecular forces M3 bromine has a melting point below room temperature (and pressure)	ACCEPT ORA throughout for iodine ACCEPT smaller atom / smaller molecule / fewer subatomic particles ALLOW less energy needed to separate molecules If no other mark scored then award 1 mark for melting point increases down the group	3
			Total 6

Question number	Answer	Notes	Marks			
4 (a)	<table border="1"><tr><td>C</td><td>yes</td><td>no</td></tr></table> A is incorrect as molten ionic compounds can be electrolysed B is incorrect as molten ionic compounds can be electrolysed but solid ionic compounds cannot D is incorrect as solid ionic compounds cannot be electrolysed	C	yes	no		1
C	yes	no				
(b) (i)	 M1 power supply connected to two electrodes M2 electrodes partially submerged in copper(II) sulfate solution in a beaker (or similar)	ACCEPT alternative symbols for power supply or labelled component OR electrodes / power supply labelled with charges instead of labels ACCEPT copper sulfate/ solution / electrolyte Max 1 mark for a short circuit or incomplete circuit	2			
(ii)	negative electrode pink-brown solid/metal positive electrode bubbles (of colourless gas)	ACCEPT effervescence	2			
(iii)	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	IGNORE state symbols	1			
			Total 6			

Question number	Answer	Notes	Marks
5 (a)	M1 (mass of CO₂ lost =) 3.1 - 1.4 OR 1.7 (g) M2 (mean rate of reaction = 1.7 ÷ 20 =) 0.085 (g/s)	ECF from M1	2
(b) (i)	 M1 all points plotted correctly	+/- ½ small square	1
(ii)	M1 best fit curve		1
(iii)	M1 decrease in rate of reaction (with larger chips) M2 (because) smaller surface area M3 (so) less frequent collisions between particles / fewer collisions between particles per unit time	ACCEPT fewer particles exposed / available at surface REJECT references to energy of the particles	3
(c)	 M1 curved graph which starts at (0,0) and has a less steep gradient M2 graph becomes horizontal at 2.5g	correct sketch scores 2 marks REJECT if graph crosses over line of the original graph	2
Total 9			

Question number	Answer	Notes	Marks
6 (a)	C (C_2H_3Cl) <i>A CHCl is incorrect</i> <i>B $C_2H_2Cl_2$ is incorrect</i> <i>D $C_4H_6Cl_2$ is incorrect</i>		1
(b) (i)	monomer(s)	ACCEPT phonetic spelling	1
(ii)			1
(c) (i)	M1 (amount of HCl =) 4 (mol) M2 (amount of PVC =) 0.00167 (mol) M3 (mass of PVC = $0.00167 \times 150\,000 =$) 250 (g) alternative method M1 (mass of 2400 mol HCl =) 87600 (g) M2 (ratio of masses =) 1:600 M3 (mass of PVC = $0.00167 \times 150\,000 =$) 250 (g)	award 3 marks for correct answer without working ACCEPT $146 \div 36.5$ ACCEPT $4 \div 2400$ ECF from M1 ACCEPT 251, 250.5 ECF from M2	3
(ii)	(hydrogen chloride gas is) toxic / acidic / corrosive	ACCEPT any named problem associated with the hazard	1
(iii)	M1 (PVC) will not react with water M2 Any one of the following: - pipe will not be damaged - pipe will last a long time - PVC will not contaminate water	ALLOW other common named substance found in air, water or ground M2 dependent on M1 ALLOW PVC will not pollute soil or (surrounding) environment	2
			Total 9

Question number	Answer	Notes	Marks
7 (a) (i)	ethene		1
(ii)	M1 (pressure) 60 - 70 atm/atmosphere(s) (inclusive) M2 (catalyst) phosphoric acid	ACCEPT any pressure between 60 and 70 inclusive ACCEPT any correct alternative unit and quantity ALLOW sulfuric acid /H ₂ SO ₄	2
(b) (i)	anaerobic / no O ₂ / absence of air	IGNORE respiration	1
(ii)	M1 enzymes become inactive below 30°C M2 yeast cells/enzymes denature above 40°C	ACCEPT ideas of rate of reaction/fermentation being too slow below 30°C If M1 and M2 not scored, (yeast contains) enzymes need an optimum temperature scores 1 mark	2
(c) (i)	 M1 for ester link M2 for the rest of the molecule with all bonds shown		2
(ii)	M1 carboxylic acid ethanoic acid M2 alcohol butanol	M2 dep on M1 ACCEPT acetic acid ACCEPT butan-1-ol/ 1-butanol	2
			Total 10

Question number	Answer	Notes	Marks
8 (a) (i)	argon	ACCEPT Ar	1
(ii)	global warming / climate change	ACCEPT any problem associated with climate change	1
(iii)	C (120 cm ³) <i>A 80 cm³ is incorrect as this is 53% of 150 cm³</i> <i>B 100 cm³ is incorrect as this is 67% of 150 cm³</i> <i>D 140 cm³ is incorrect as this is 93% of 150 cm³</i>	ALLOW greenhouse gas	1
(b)	M1 sulfur combusts/reacts with oxygen (in the air) M2 $S + O_2 \rightarrow SO_2$ M3 the (amount/volume of) oxygen is replaced with the (same amount/volume of) sulfur dioxide		3
			Total 6

Question number	Answer	Notes	Marks
9 (a)	(i) white precipitate		1
	(ii) $\text{CuCl}_2(\text{aq}) + \text{BaSO}_4(\text{s})$ M1 formulae of products M2 state symbols	ACCEPT formulae in any order M2 dependent on M1	2
(b)	M1 dip a metal / nichrome wire (into hydrochloric acid and then) into the solid M2 place (the wire and solid) into a non-luminous/roaring Bunsen burner flame M3 blue-green flame	ALLOW silica rod ACCEPT idea of non-luminous Bunsen burner flame	3
(c)	(i) M1 heat the liquid/water M2 (confirm) the boiling point is 100°C OR M1 cool the liquid/water M2 (confirm) the freezing / melting point is 0°C	ACCEPT boil the liquid/water ACCEPT (pure) water boils at 100°C ACCEPT freeze the liquid/water ACCEPT (pure) water freezes at 0°C	2
	(ii) M1 add water / the colourless liquid (to the anhydrous copper(II) sulfate) M2 turns (back to) blue / turns (from white/colourless) to blue	ACCEPT addition of substances the other way around ACCEPT changes back to original colour ALLOW heat given off / temperature increase M2 dependent on M1	2
	(iii) hydrated copper(II) sulfate $\rightleftharpoons$ anhydrous copper(II) sulfate + water	ALLOW equation written the other way around	1

(d)	M1 (mass of $\text{CuSO}_4 =$) 3.99 (g) M2 (mass of $\text{H}_2\text{O} =$) 2.25 (g) M3 (amount of $\text{CuSO}_4 =$) 0.025 (mol) M4 (amount of $\text{H}_2\text{O} =$) 0.125 (mol) M5 value of $x = 0.125 \div 0.025 (= 5)$ alternative method M1 (mass of $\text{CuSO}_4 =$) 3.99 (g) M2 (mass of hydrated copper sulfate =) 6.24 (g) M3 (M_r of hydrated copper sulfate =) 249.5 M4 (M_r of $x\text{H}_2\text{O} =$) 90 M5 value of $x = 90 \div 18 (= 5)$	ACCEPT 24.49 - 20.50 ACCEPT 26.74 - 24.49 ACCEPT $3.99 \div 159.5$ ACCEPT $2.25 \div 18$ ACCEPT 24.49 - 20.50 ACCEPT 26.74 - 20.50 ACCEPT $159.5 \div (3.99 \times 6.24)$ ACCEPT 249.5 - 159.5	5
			Total 16

Question number	Answer	Notes	Marks
10 (a) (i)	(amount of product at equilibrium) > (amount of reactants at equilibrium) (rate of forwards reaction at equilibrium) = (rate of reverse reaction at equilibrium)		1
(ii)	M1 the reaction is not being catalysed M2 (so) activation energy is no longer lowered (by an alternative pathway) M3 (more) molecules do not have (energy equal to or greater than) the activation energy OR reaction requires more energy	ALLOW iron is a catalyst ACCEPT activation energy increases ACCEPT fewer molecules have the energy required to react ACCEPT particles for molecules	3
(b)	M1 (amount of nitrogen =) 182 (mol) M2 (theoretical amount of ammonia =) 364 (mol) M3 (actual amount of ammonia =) 255 (mol) M4 (minimum mass of ammonia = 255×17 =) 4300 (g) alternative method M1 (amount of nitrogen =) 182 (mol) M2 (theoretical amount of ammonia =) 364 (mol) M3 (theoretical mass of ammonia =) 6188 (g) M4 (minimum mass of ammonia = $6188 \times 70 \div 100$ =) 4300 (g)	Correct answer without working scores 4 marks ACCEPT percentage calculation at any step of working ACCEPT $5100 \div 28$ ECF from M1 ACCEPT 182×2 ECF from M2 ACCEPT $364 \times 70 \div 100$ ECF from M3 ACCEPT 4335 OR 4340 ACCEPT $5100 \div 28$ ECF from M1 ACCEPT 182×2 ECF from M2 ACCEPT 364×17 ECF from M3 ACCEPT 4332 OR 4330	4

(c)	M1 (both new conditions cause) decrease in percentage yield (of ammonia) M2 (because the position of) equilibrium shifts backwards / to the left hand side (due to the change in both conditions) M3 (the condition of) pressure has been decreased M4 (which) favours the side (of equation) with more molecules / moles (of gas) M5 (the condition of) temperature has been increased M6 (which) favours the reaction (direction) that is endothermic	ACCEPT reaction/equation moves backwards/to the left	6
			Total 14